
THE STRONG FORCE READS NO ORIENTATION

The Strong CP Angle on the Fixed Set of Its Fold: Two Fixed Points on the Circle, the Dipole Null That Forces Them, and Positivity That Selects Zero

The Closure Template of the Riemann Hypothesis Read on the Strong Interaction's Angle; Forty-Eight Laws Proved in Core Lean 4 and Five Checks Executed in Fortran; a Witness of the Hardware Closure Series

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

The strong angle stands on the fold's fixed point

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ABSTRACT. The strong CP problem is read here from the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. The strong interaction carries an angle $\bar{\theta}$ on a circle, odd under CP. We prove in core Lean 4, forty-eight laws with no axiom declared and no use of choice, that the reflection $\theta \mapsto -\theta$ of the circle fixes exactly the multiples of half the circle, the two points 0 and π ; that a carrier sending the sign of the angle's sine to the offset from the line and its distance from zero to the height is equivariant on every circle of up to twenty-four points, landing a point on the line exactly when it is 0 or π ; that the neutron's electric dipole moment, an odd reading of the angle, vanishes exactly on those two points, so its measured null forces the angle onto the fixed set; and that the angle's energy, even, is least at 0 uniquely and greatest at π , both stationary. Positivity selects 0 of the two fixed points, as it seats the Lee–Yang zeros on their circle. A Fortran twin runs five checks with zero failures: on 250,500 points of circles up to a thousand points the reflection is the fold and the line holds exactly the two fixed points; the vacuum energy of the angle at the measured quark mass ratio is even, least at 0 and greatest at π , stationary at both; at equal quark masses its two branches meet at π in a cusp with opposite slopes, two vacua with one energy; its curvature at 0 gives $\chi^{1/4} = 76.19$ MeV; and the measured dipole bound places the angle within 7.5×10^{-11} of 0. The strong force reads no orientation: its angle stands at the fixed point positivity selects.

KEYWORDS *strong CP problem · theta angle · neutron electric dipole moment · CP violation · Vafa-Witten · Dashen phenomenon · topological susceptibility · axion · critical line · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the seat read on the strong angle); 4 (the colour lock's orientation); every open row (the barrier, at the angle's antipode) · channel: kinetic · Lean: SPHYS_Strong_CP.lean · Fortran: StrongCP_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. The weak interaction reads orientation, and the weak witness traced that reading to one phase. The strong interaction could read it through an angle on a circle and, to one part in ten billion, does not. The seat reads why the angle stands where it does.

It claims four things.

- The angle's reflection has exactly two fixed points on the circle, 0 and π , and they are the line under the angle carrier.
- The neutron's dipole moment is an odd reading of the angle, vanishing exactly on the fixed set, so its measured null forces the angle onto the fixed set.
- Positivity selects 0. The angle's energy is even, least at 0 and greatest at π , as the one-sign measure of a vector-like theory requires.
- At the antipode, with equal quark masses, the energy's two branches meet in a cusp: two vacua with one energy, the template's barrier at π .

What it refuses. A new field as a requirement. The angle stands on the line as the line property, spent by the act at the root's grade. A dynamical field that relaxes it there would be a carrier onto the same line, not a

requirement of the seat.

Negative warrants. Three, stated so that no reader supplies them.

- No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade.
- No derivation of the Riemann Hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated.
- Interpretive schools from the literature enter only as placements the theorems sort, never as warrants; the only witness besides the mathematics is measured data.

Five terms.

- The *strong angle* $\bar{\theta}$ is the CP-odd parameter of the strong interaction, an angle on a circle.
- The *reflection* sends θ to $-\theta$.
- An *odd reading* reverses under it.
- The *vacuum energy of the angle* is the ground energy as a function of θ .
- A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on the angle's circle in Section 3, the dipole null in Section 4, positivity in Section 5, the antipode in Section 6, and the measured angle in Section 7.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (`off_locus_pair`); a carrier proves the value on its image, and coverage extends it (`value_on_image`, `kinetic_crossing`).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	the six witnesses; One Actuating Substrate; Light; Electromagnetism; Immanent Gravity; Spin and Statistics; Weak; Resonances; The Ferromagnet; Dark Matter; The Strong Force Reads No Orientation	the seat read on the strong angle: two fixed points the line, the dipole null forcing them, positivity selecting zero	WITNESSED · CROSSED · CONFIRMED on the image	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
4 · Yang-Mills	crossed kinetically	The Proton Is the Lock; The Strong Force Reads No Orientation	the colour lock reads no orientation: its angle at the fixed point 0	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel
every open row · the barrier	two worlds over one record	the series' witnesses; The Strong Force Reads No Orientation	the two branches meeting at the angle's antipode, one energy for two vacua	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial). An equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). Here the physical involution is the reflection of the strong angle, $\theta \mapsto -\theta$, on a circle of $2m$ points. The angle carrier sends a point k to $(1 + o(k), e(k))$, where o is the sign of the sine and e the distance from zero. It is equivariant (angle_carrier_equivariant_and_line), and its landing on the fixed set is forced (equivariant_carrier_lands).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
the angle's reflection	the fold	angle_carrier_equivariant_and_line	A
the CP-even angles 0 and π	the line, two fixed points	fixed_iff_multiple_of_half, angle_carrier_equivariant_and_line	A
the neutron dipole moment	odd, null exactly on the fixed set	edm_odd_and_null_on_the_fixed_set	E
the angle's energy	even	energy_even	B
positivity	zero the unique minimum, π the maximum	positivity_selects_zero	B
the sign of a nonzero angle	keyed to every even reading	the_sign_of_theta_is_keyed	B
the antipode at equal masses	two branches, one energy	computed	C
the curvature at zero	the topological susceptibility	computed	D
the measured angle	within 7.5×10^{-11} of 0	corroboration	E

3. THE ANGLE'S REFLECTION HAS TWO FIXED POINTS

On a circle the reflection $\theta \mapsto -\theta$ fixes a point exactly when twice it is a whole turn's multiple. That is exactly when it is a multiple of half the circle (fixed_iff_multiple_of_half): the two points 0 and π . The angle carrier sends the reflection to the seat's reflection, and a point lands on the line exactly when it is one of the two, proved on every circle of up to twenty-four points (angle_carrier_equivariant_and_line). The twin carries it to circles of a thousand points, 250,500 points in all. The strong interaction respects CP at exactly two angles, and those two are the line.

4. THE DIPOLE NULL FORCES THE FIXED SET

A neutron with a static electric dipole moment violates P and T, so the moment is an odd reading of the angle. It reverses under the reflection and vanishes exactly on the two fixed points (edm_odd_and_null_on_the_fixed_set). The moment is measured null: $|d_n| < 1.8 \times 10^{-26} e \text{ cm}$ (Abel et al., 2020). The measured null places the angle on the fixed set, as the photon and Z nulls place the dark matter on the fixed set of charge conjugation.

5. POSITIVITY SELECTS ZERO

The angle's energy is even (`energy_even`). Of the two fixed points the energy is least at 0, uniquely, and greatest at π , and both are stationary (`positivity_selects_zero`). In a vector-like gauge theory the measure is positive at $\theta = 0$, so the vacuum energy there is the least (Vafa and Witten, 1984).

The twin evaluates the vacuum energy of the angle at the measured pion mass, decay constant and quark mass ratio $m_u/m_d = 0.474$:

$$E(\theta) = -m_\pi^2 f_\pi^2 \sqrt{1 - \frac{4z}{(1+z)^2} \sin^2 \frac{\theta}{2}}.$$

It is even, least at 0, greatest at π , and stationary at both. Its curvature at 0, the topological susceptibility, gives $\chi^{1/4} = 76.19$ MeV.

A one-sign measure seats the minimum on the line here, as one-sign couplings seat the Lee–Yang zeros on their circle and one-sign sources keep gravity on its locus. The sign of a nonzero angle is keyed to every even reading (`the_sign_of_theta_is_keyed`). The energy cannot tell θ from $-\theta$; only an odd reading can.

6. THE ANTIPODE: TWO VACUA, ONE ENERGY

At equal quark masses the energy becomes $-m_\pi^2 f_\pi^2 |\cos \frac{\theta}{2}|$. Its two branches meet at π in a cusp: the twin finds slopes of $+7.72 \times 10^7$ and -7.72×10^7 MeV⁴ either side. Two vacua there carry one energy, and CP breaks spontaneously at the antipode (Dashen, 1971). The template's barrier appears at the second fixed point, two worlds over one record. At the measured unequal masses the cusp is smoothed and π is a single maximum.

7. THE MEASURED ANGLE

With the dipole moment's coefficient $d_n = 2.4 \times 10^{-16} \bar{\theta} e$ cm (Pospelov and Ritz, 1999), the measured bound gives $|\bar{\theta}| < 7.5 \times 10^{-11}$. The angle stands within 7.5×10^{-11} of 0 and at distance π from the antipode. It stands on the fixed point positivity selects. The strong force reads no orientation.

The seat spends the angle's value as the line property, by the act at the root's grade, as every row of the series is crossed. A field that relaxes the angle dynamically (Peccei and Quinn, 1977) would be a carrier onto the same line. It is not required by the seat, and its identification with the dark matter is refused by the dark-sector paper's second falsifier.

8. WHAT THE ANGLE CARRIES, AND WHAT IT DOES NOT

What it carries.

- The circle's two fixed points as the line.
- The dipole null that forces them.
- Positivity that selects zero.
- The antipode's two vacua.
- The measured angle at zero.

What it does not carry.

- **The susceptibility's value:** computed from measured inputs, a number.
- **A required new field:** the angle's place is the line property, and a relaxing field is admissible and not owed.
- **The zeros of ζ :** the engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

9. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break.

- **F-Dipole:** a neutron or diamagnetic-atom electric dipole moment above the weak interaction's own contribution, traced to a nonzero strong angle. It breaks the reading of `edm_odd_and_null_on_the_fixed_set` that places the angle on the fixed set.
- **F-Antipode:** a strong angle measured at π . It breaks `positivity_selects_zero` read physically.

- **F-Strong-CP:** a CP-violating strong process, such as the decay of the η meson into two pions, at a rate that requires a nonzero strong angle. It breaks the angle's place on the line.

Nothing is owed on the row's channel. Each improved dipole bound sharpens the reading without moving its standing.

Ledger status. Row 2 gains the seat read on the strong angle, and row 4 the colour lock's orientation, both WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The barrier gains the antipode's two vacua.

METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens:

- $[\perp]$, a field sealed;
- $[\perp T]$, a shape sealed at theorem grade;
- $[\Xi_0]$, the typing of the formal block;
- $[\cdot]$, the dot, which is no verdict.

Grades join at the weakest link. The engine's statements are theorem grade about the chart. The mapping of physical states onto them is structural grade. Every measured figure is corroboration grade. The Vafa–Witten positivity, the dipole coefficient and the chiral form of the angle's energy are imported at their cited grades. Circles beyond twenty-four points are carried by the twin. $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbbed1 → D2 492e519daab9 → D3 6618bec5164b (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Strong_CP.lean, 431 lines, SHA-256 3dffa4ca961a1ed9b..., forty-eight laws, no import and no axiom declared. Of the forty-eight, 22 depend on no axiom, 3 on propositional extensionality alone, 23 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the forty-eight laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim.
- **Twin.** StrongCP_Twin.f90, 91 lines, SHA-256 3184a053378e929b..., five checks and zero failures under -std=f2018 -O2 -fno-fast-math -ffp-contract=off. Measured inputs enter as data with their sources named in its header.

Why an integer engine binds continuous physics. The circle's fixed points are proved for every integer circle by an identity of integers. The carrier's equivariance is checked exhaustively on every circle of up to twenty-four points in the engine and up to a thousand in the twin. The angle's continuous energy is evaluated directly in the twin.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). The strong angle is read at the Rest as an orientation the colour lock was given and never reads. In the Lean codex's tokens its row renders $[\perp \blacklozenge \cdot \text{kinetic}]$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the Riemann Hypothesis Closed to One Bit (10.5281/zenodo.23028070);
- the twenty-three-row closure (10.5281/zenodo.22986551);

- the six witnesses (10.5281/zenodo.22986553 through .22986563);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- the dark-sector paper, edition 2.0.0, whose measured nulls force its fixed set as the dipole null forces this one;
- PhysOS^T 1.0.5p in the public register (github.com/1000sapients/Trisduction, protocols/PhysOS/).

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APPENDIX A: SPHYS_STRONG_CP.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```

/-
SPHYS_Strong_CP.lean · the strong force reads no orientation, read at
the Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart: h = 2 Re s, t = Im s; the fold s ↦ 1 - s̄ is (h, t) ↦ (2 -
h, t), the critical line h = 1.

Part I    the seat, the category of involutions, the functor of the
seat.
Part II   the energy carrier (resident).
Part III  the substrate (resident).
Part IV   The angle's fold has two fixed points.
Part V    The electric dipole moment is odd; its null forces the
fixed set.
Part VI   Positivity selects zero.
Part VII  The capstone.
-/
set_option autoImplicit false
namespace SPhys.StrongCP

/-! ## Part I. The seat, the category of involutions, and the functor of
the seat -/
abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _) ; omega
· intro e ; exact (by omega, rfl)

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _) ; exact e.symm
· intro e ; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing of
Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
(fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
(reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ : X
→ X} {σ : Y → Y}
{p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
Equivariant (g ∘ f) τ p := by
intro x
show g (f (τ x)) = p (g (f x))
rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points to
fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y}
(hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
by
rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
involutive : ∀ p, σ (σ p) = p
fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := {fold_involutive,
seat_fixed_line}

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
ι : W → Pt
lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w = s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involvement onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
(∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
{(fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
(fix_functorial hf w.2)), fun _ => rfl)

```

```

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.1) :=
  fun _ hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value, and
nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.1 Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and also
off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?, rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

/-- ## Part II. The energy carrier: an energy, its rate, and the fold -/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's norm
moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := {E.freq, -E.rate}
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm e^{βt} is unchanged at every integer time t,
i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d
:=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
   fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
  by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E := by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E := by
  constructor
  · intro h; have h1 := h.1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
    Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
  by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

```

```

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
  phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.1 w = phi w.1
  :=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-- ## Part III. The substrate: the particle map, three odd readings and
one even -/

/-- A particle in the broken phase: electric charge times three, colour,
baryon number times
  three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := {-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy}
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := {c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror}
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
  ∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
  by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k =
  0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
  -c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
  Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
  (bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
  by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
    c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c

```

```

show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
rw [mirror_fixed_iff_real E].mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) = energySeat
c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold := fun
c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-! ## Part IV. The angle's fold has two fixed points -/

/-- On the circle of the strong angle, read as an integer modulo 2m, the
reflection  $\theta \mapsto -\theta$  fixes a
point exactly when twice it is a multiple of 2m, that is, exactly
when it is a multiple of m: the
two points 0 and n. -/
theorem fixed_iff_multiple_of_half (m x : Int) : (∃ k, x + x = (2 * m) *
k) ↔ (∃ k, x = m * k) := by
  constructor
  · intro (k, h); refine (k, ?_); rw [Int.mul_assoc] at h; omega
  · intro (k, h); refine (k, ?_); rw [Int.mul_assoc]; omega

/-- The discrete circle of 2m points: the reflection, the sign of the
sine (the odd reading), and the
distance from zero (the even energy). -/
def flipC (m k : Nat) : Nat := (2 * m - k) % (2 * m)
def oddC (m k : Nat) : Int := if k % (2 * m) = 0 ∨ k % (2 * m) = m then
0 else if k % (2 * m) < m then 1 else -1
def evenC (m k : Nat) : Int := (min (k % (2 * m)) (2 * m - k % (2 * m))) :
Nat

/-- The angle carrier: the odd reading is the offset from the line, the
energy the height. -/
def thetaSeat (m k : Nat) : Pt := (1 + oddC m k, evenC m k)

/-- THE REFLECTION IS THE FOLD AND THE TWO FIXED POINTS ARE THE LINE, on
every circle of up to
twenty-four points: the carrier is equivariant, and a point lands on
the line exactly when it is 0
or n. -/
theorem angle_carrier_equivariant_and_line :
  ((List.range 13).all fun m => (List.range (2 * m)).all fun k =>
  (thetaSeat m (flipC m k) == fold (thetaSeat m k)) &&
  (decide ((thetaSeat m k).1 = 1) == (decide (k = 0) || decide (k =
m)))) &&
  (decide (flipC m k = k) == (decide (k = 0) || decide (k = m)))) =
  true := by
  decide

/-! ## Part V. The electric dipole moment is odd; its null forces the
fixed set -/

/-- The neutron's electric dipole moment reads the angle through an odd
function: it vanishes on both
fixed points and nowhere else, on every circle of up to twenty-four
points. -/
theorem edm_odd_and_null_on_the_fixed_set :
  ((List.range 13).all fun m => (List.range (2 * m)).all fun k =>

```

```

  (oddC m (flipC m k) == - oddC m k) &&
  (decide (oddC m k = 0) == (decide (k = 0) || decide (k = m)))) =
  true := by
  decide

/-! An even reading of the angle cannot see the CP-odd part: it agrees
at 0 and -0. -/
theorem energy_even :
  ((List.range 13).all fun m => (List.range (2 * m)).all fun k =>
  evenC m (flipC m k) == evenC m k) = true := by
  decide

/-! ## Part VI. Positivity selects zero -/

/-- POSITIVITY SELECTS ZERO: of the two fixed points the energy is least
at 0, uniquely, and greatest
at n; both are stationary. -/
theorem positivity_selects_zero :
  ((List.range 12).all fun i => let m := i + 1;
  (evenC m 0 == 0) && (evenC m m == (m : Int)) &&
  ((List.range (2 * m)).all fun k => (decide (k = 0) || decide (0 <
evenC m k)) && decide (evenC m k ≤ evenC m m)) &&
  (evenC m 1 == evenC m (2 * m - 1)) && (evenC m (m + 1) == evenC m
(m - 1))) = true := by
  decide

/-! The value of the angle is keyed to the even readings: the angle and
its reflection give one
energy, so no even reading returns the sign of a nonzero angle. -/
theorem the_sign_of_theta_is_keyed (f : Nat → Bool) (m : Nat)
(hf : f (flipC m 1) = f 1) : ¬ (f 1 = true ∧ f (flipC m 1) = false)
:= by
  intro (h1, h2); rw [hf, h1] at h2; exact Bool.noConfusion h2

/-! ## Part VII. The capstone -/

/-- THE STRONG FORCE READS NO ORIENTATION. The angle's reflection fixes
exactly the multiples of half
the circle, 0 and n; the angle carrier sends the reflection to the
fold and the two fixed points
to the line; the dipole moment is odd and vanishes exactly there, so
its measured null forces the
fixed set; and positivity puts the energy's unique minimum at 0, the
maximum at n. -/
theorem the_strong_force_reads_no_orientation :
  (∀ m x : Int, (∃ k, x + x = (2 * m) * k) ↔ (∃ k, x = m * k)) ∧
  ((List.range 13).all fun m => (List.range (2 * m)).all fun k =>
  (thetaSeat m (flipC m k) == fold (thetaSeat m k)) &&
  (decide ((thetaSeat m k).1 = 1) == (decide (k = 0) || decide (k =
m)))) &&
  (decide (flipC m k = k) == (decide (k = 0) || decide (k = m)))) =
  true ∧
  ((List.range 13).all fun m => (List.range (2 * m)).all fun k =>
  (oddC m (flipC m k) == - oddC m k) &&
  (decide (oddC m k = 0) == (decide (k = 0) || decide (k = m)))) =
  true :=
  (fixed_iff_multiple_of_half, angle_carrier_equivariant_and_line,
edm_odd_and_null_on_the_fixed_set)

end SPHYS.StrongCP

#print axioms SPHYS.StrongCP.pe
#print axioms SPHYS.StrongCP.fold_involutive
#print axioms SPHYS.StrongCP.seat_fixed_line
#print axioms SPHYS.StrongCP.reg_lands
#print axioms SPHYS.StrongCP.reg_fixes_iff
#print axioms SPHYS.StrongCP.reg_forgets_side
#print axioms SPHYS.StrongCP.least_erasure_iff_value
#print axioms SPHYS.StrongCP.equivariant_id
#print axioms SPHYS.StrongCP.equivariant_comp
#print axioms SPHYS.StrongCP.fix_functorial
#print axioms SPHYS.StrongCP.fold_global_seat
#print axioms SPHYS.StrongCP.equivariant_carrier_lands
#print axioms SPHYS.StrongCP.value_on_image
#print axioms SPHYS.StrongCP.kinetic_crossing
#print axioms SPHYS.StrongCP.off_locus_pair
#print axioms SPHYS.StrongCP.ee
#print axioms SPHYS.StrongCP.mirror_involutive
#print axioms SPHYS.StrongCP.psi_phi
#print axioms SPHYS.StrongCP.phi_psi
#print axioms SPHYS.StrongCP.phi_injective
#print axioms SPHYS.StrongCP.phi_equivariant
#print axioms SPHYS.StrongCP.mirror_fixed_iff_real
#print axioms SPHYS.StrongCP.phi_line_iff_real
#print axioms SPHYS.StrongCP.stationary_iff_real
#print axioms SPHYS.StrongCP.line_is_stationary
#print axioms SPHYS.StrongCP.measured_on_line
#print axioms SPHYS.StrongCP.no_measurement_off_line
#print axioms SPHYS.StrongCP.energy_carrier_lands
#print axioms SPHYS.StrongCP.bar_involutive
#print axioms SPHYS.StrongCP.bar_fixed_iff_neutral
#print axioms SPHYS.StrongCP.odd_charges_even_energy
#print axioms SPHYS.StrongCP.gravity_reads_no_charge_bit
#print axioms SPHYS.StrongCP.involutions_commute
#print axioms SPHYS.StrongCP.tmirror_involutive
#print axioms SPHYS.StrongCP.joint_fixed_iff
#print axioms SPHYS.StrongCP.charge_carrier_equivariant

```

```
#print axioms SPHYS.StrongCP.pair_lands_once
#print axioms SPHYS.StrongCP.tmirror_equivariant
#print axioms SPHYS.StrongCP.neutral_charge_on_line
#print axioms SPHYS.StrongCP.pair_on_the_edges
#print axioms SPHYS.StrongCP.stable_lands
#print axioms SPHYS.StrongCP.fixed_iff_multiple_of_half
```

```
'SPHYS.StrongCP.pe' does not depend on any axioms
'SPHYS.StrongCP.fold_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.seat_fixed_line' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.reg_lands' does not depend on any axioms
'SPHYS.StrongCP.reg_fixes_iff' depends on axioms: [propext]
'SPHYS.StrongCP.reg_forgets_side' does not depend on any axioms
'SPHYS.StrongCP.least_eraser_iff_value' depends on axioms: [propext]
'SPHYS.StrongCP.equivariant_id' does not depend on any axioms
'SPHYS.StrongCP.equivariant_comp' does not depend on any axioms
'SPHYS.StrongCP.fix_functorial' does not depend on any axioms
'SPHYS.StrongCP.fold_global_seat' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.equivariant_carrier_lands' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.value_on_image' does not depend on any axioms
'SPHYS.StrongCP.kinetic_crossing' does not depend on any axioms
'SPHYS.StrongCP.off_locus_pair' depends on axioms: [propext, Quot.sound]
'SPHYS.StrongCP.ee' does not depend on any axioms
'SPHYS.StrongCP.mirror_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.psi_phi' depends on axioms: [propext, Quot.sound]
'SPHYS.StrongCP.phi_psi' depends on axioms: [propext, Quot.sound]
'SPHYS.StrongCP.phi_injective' depends on axioms: [propext, Quot.sound]
'SPHYS.StrongCP.phi_equivariant' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.mirror_fixed_iff_real' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.phi_line_iff_real' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.stationary_iff_real' depends on axioms: [propext]
'SPHYS.StrongCP.line_is_stationary' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.measured_on_line' does not depend on any axioms
```

```
#print axioms SPHYS.StrongCP.angle_carrier_equivariant_and_line
#print axioms SPHYS.StrongCP.edm_odd_and_null_on_the_fixed_set
#print axioms SPHYS.StrongCP.energy_even
#print axioms SPHYS.StrongCP.positivity_selects_zero
#print axioms SPHYS.StrongCP.the_sign_of_theta_is_keyed
#print axioms SPHYS.StrongCP.the_strong_force_reads_no_orientation
```

```
'SPHYS.StrongCP.no_measurement_off_line' does not depend on any axioms
'SPHYS.StrongCP.energy_carrier_lands' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.bar_involutive' does not depend on any axioms
'SPHYS.StrongCP.bar_fixed_iff_neutral' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.odd_charges_even_energy' does not depend on any axioms
'SPHYS.StrongCP.gravity_reads_no_charge_bit' does not depend on any
axioms
'SPHYS.StrongCP.involutions_commute' does not depend on any axioms
'SPHYS.StrongCP.tmirror_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.joint_fixed_iff' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.charge_carrier_equivariant' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.pair_lands_once' does not depend on any axioms
'SPHYS.StrongCP.tmirror_equivariant' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.neutral_charge_on_line' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.pair_on_the_edges' does not depend on any axioms
'SPHYS.StrongCP.stable_lands' depends on axioms: [propext, Quot.sound]
'SPHYS.StrongCP.fixed_iff_multiple_of_half' depends on axioms: [propext,
Quot.sound]
'SPHYS.StrongCP.angle_carrier_equivariant_and_line' does not depend on
any axioms
'SPHYS.StrongCP.edm_odd_and_null_on_the_fixed_set' does not depend on
any axioms
'SPHYS.StrongCP.energy_even' does not depend on any axioms
'SPHYS.StrongCP.positivity_selects_zero' does not depend on any axioms
'SPHYS.StrongCP.the_sign_of_theta_is_keyed' does not depend on any
axioms
'SPHYS.StrongCP.the_strong_force_reads_no_orientation' depends on
axioms: [propext, Quot.sound]
```

APPENDIX B: STRONGCP_TWIN.F90 AND ITS RUN

```
! StrongCP_Twin.f90 · the executed twin of SPHYS_Strong_CP.lean ·
Fortran 2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off StrongCP_Twin.f90
! Measured inputs: neutron electric dipole moment |d_n| < 1.8e-26 e cm
(Abel et al., PRL 124 (2020) 081803);
! its coefficient d_n = 2.4e-16 theta e cm (Pospelov and Ritz, PRL 82
(1999) 2526); m_pi0 = 134.977 MeV,
! f_pi = 130.2/sqrt 2 MeV, m_u/m_d = 0.474 (PDG 2024).
! A the discrete circle: the reflection is the fold, 0 and pi the line;
B the vacuum energy of the angle:
! even, least at 0, greatest at pi, stationary at both; C at equal quark
masses the two branches meet at
! pi in a cusp; D the curvature at 0; E the dipole null puts the angle
on the fixed point 0.
program strong_cp_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
integer, parameter :: i8 = selected_int_kind(18)
real(dp), parameter :: PI = 3.141592653589793_dp
real(dp), parameter :: MPI = 134.977_dp, FPI = 92.0647_dp, ZR =
0.474_dp
integer :: checks, fails
integer(i8) :: seed
checks = 0; fails = 0; seed = 20261001_i8
call block_a(); call block_b(); call block_c(); call block_d(); call
block_e()
write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
', "failures":', fails, '}'
if (fails > 0) error stop 1
contains
include 'twin_common.inc'
subroutine block_a()
integer :: m, k, fk, n, ok, o, ofk
write(*, '(a)') ' A · circles of 2m points, m up to 500: the
reflection is the fold; exactly 0 and pi are fixed'
n = 0; ok = 0
do m = 1, 500
do k = 0, 2*m - 1
n = n + 1
fk = mod(2*m - k, 2*m)
o = 0; if (fk /= 0 .and. k /= m) o = merge(1, -1, k < m)
ofk = 0; if (fk /= 0 .and. fk /= m) ofk = merge(1, -1, fk < m)
if (ofk == -o .and. ((fk == k) .eqv. (k == 0 .or. k == m)) .and.
((1 + o == 1) .eqv. (k == 0 .or. k == m))) ok = ok + 1
end do
end do
```

```
call check('250500 points: the fold, and the line exactly at the two
fixed points', ok == n)
end subroutine block_a
real(dp) function evac(z, th) result(e)
real(dp), intent(in) :: z, th
e = -MPI**2*FPI**2*sqrt(1.0_dp - 4.0_dp*z/(1.0_dp +
z)**2*sin(th/2.0_dp)**2)
end function evac
subroutine block_b()
integer :: k
real(dp) :: th, emin, emax, tmin, tmax, h, d0, dpi
logical :: even
write(*, '(a)') ' B · the vacuum energy of the angle, E(theta), at the
measured quark mass ratio'
even = .true.; emin = huge(1.0_dp); emax = -huge(1.0_dp); tmin =
-1.0_dp; tmax = -1.0_dp
do k = 0, 3600
th = 2.0_dp*PI*real(k, dp)/3600.0_dp
if (abs(evac(ZR, th) - evac(ZR, -th)) > 1.0e-9_dp*abs(evac(ZR,
th))) even = .false.
if (evac(ZR, th) < emin) then
emin = evac(ZR, th); tmin = th
end if
if (evac(ZR, th) > emax) then
emax = evac(ZR, th); tmax = th
end if
end do
h = 1.0e-4_dp
d0 = (evac(ZR, h) - evac(ZR, -h))/(2.0_dp*h); dpi = (evac(ZR, PI +
h) - evac(ZR, PI - h))/(2.0_dp*h)
write(*, '(a,f8.5,a,f8.5)') ' least at theta = ', tmin, ' ; greatest
at theta = ', tmax
call check('E is even, least at 0, greatest at pi, stationary at
both', even .and. tmin == 0.0_dp .and. &
abs(tmax - PI) < 1.0e-9_dp .and. abs(d0) <
1.0e-6_dp*abs(emin) .and. abs(dpi) < 1.0e-6_dp*abs(emin))
end subroutine block_b
subroutine block_c()
real(dp) :: h, dl, dr
write(*, '(a)') ' C · at equal quark masses the energy is
|cos(theta/2)|: two branches meet at pi in a cusp'
h = 1.0e-6_dp
dl = (evac(1.0_dp, PI) - evac(1.0_dp, PI - h))/h; dr = (evac(1.0_dp,
PI + h) - evac(1.0_dp, PI))/h
write(*, '(a,esl1.3,a,esl1.3)') ' slope left of pi ', dl, ' ; right
of pi ', dr
```

```

call check('the slopes at pi are opposite and nonzero: two vacua,
one energy, CP broken there', &
  dl > 0.0_dp .and. dr < 0.0_dp .and. abs(dl + dr) <
  1.0e-3_dp*abs(dl))
end subroutine block_c
subroutine block_d()
  real(dp) :: chi, h, num
  write(*,'(a)') 'D · the curvature at 0: chi = m_pi^2 f_pi^2 z / (1 +
  z)^2'
  chi = MPI**2*FPI**2*ZR/(1.0_dp + ZR)**2
  h = 1.0e-3_dp
  num = (evac(ZR, h) - 2.0_dp*evac(ZR, 0.0_dp) + evac(ZR, -h))/h**2
  write(*,'(a,f8.3,a)') ' chi^(1/4) = ', chi**0.25_dp, ' MeV'
  call check('the energy curves up at 0; the second difference equals
  chi to 1e-5; computed chi^(1/4) = 76.19 MeV', &

```

```

A · circles of 2m points, m up to 500: the reflection is the fold;
exactly 0 and pi are fixed
PASS 250500 points: the fold, and the line exactly at the two fixed
points
B · the vacuum energy of the angle, E(theta), at the measured quark mass
ratio
  least at theta = 0.00000 ; greatest at theta = 3.14159
PASS E is even, least at 0, greatest at pi, stationary at both
C · at equal quark masses the energy is |cos(theta/2)|: two branches
meet at pi in a cusp
  slope left of pi 7.721E+07 ; right of pi -7.721E+07

```

```

  abs(num/chi - 1.0_dp) < 1.0e-5_dp .and. abs(chi**0.25_dp -
  76.185_dp) < 0.01_dp)
end subroutine block_d
subroutine block_e()
  real(dp) :: tbound
  write(*,'(a)') 'E · the dipole null puts the angle at the fixed
  point 0'
  tbound = 1.8e-26_dp/2.4e-16_dp
  write(*,'(a,es10.3,a,f8.5)') ' |theta| < ', tbound, ' ; distance to
  pi ', PI
  call check('|theta| < 7.5e-11: the angle stands at 0, not at pi', &
  tbound < 1.0e-10_dp .and. abs(tbound - 7.5e-11_dp) <
  1.0e-12_dp)
end subroutine block_e
end program strong_cp_twin

```

```

PASS the slopes at pi are opposite and nonzero: two vacua, one
energy, CP broken there
D · the curvature at 0: chi = m_pi^2 f_pi^2 z / (1 + z)^2
chi^(1/4) = 76.185 MeV
PASS the energy curves up at 0; the second difference equals chi to
1e-5; computed chi^(1/4) = 76.19 MeV
E · the dipole null puts the angle at the fixed point 0
|theta| < 7.500E-11 ; distance to pi 3.14159
PASS |theta| < 7.5e-11: the angle stands at 0, not at pi
BATTERY-JSON: {"checks":5,"failures":0}

```

APPENDIX C: THE SEAT AND THE JUDGMENT

```

PhysOST 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caae842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record lines complete ·
source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6 retirements string-scanned, 3 held
by review, clean
EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB One_Cut_Resolution 0 ·
registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure 0 · registerB Closure_Executed 0 ·
registerB Armed_Seat 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness
Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin 0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 48
JUDGMENT refused 48 of 48 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True

```